CS-301: Computer Networks Lab

Cisco Packet Tracer — Assignment 4

All practicals must be completed inside **Cisco Packet Tracer**.
Each assignment requires exactly **two screenshots** as submission.

Subject: CS-301 — Computer Networks
Tool: Cisco Packet Tracer 8.x or later

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_SetX.pkt` (e.g. `20221234_Set4.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of each set.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

Sunrise Exports has just moved into a two-storey building. The ground floor is occupied by the **Sales** team and the first floor by the **Accounts** team. At the moment the two floors are completely separate — a Sales PC cannot reach an Accounts PC and vice versa, which is causing daily frustration for staff who need to share files.

The IT coordinator, Mr. Sameer Joshi, tells you that each floor already has its own switch and its own set of PCs. He simply needs the two switches **linked together** and the machines **addressed correctly** so the entire office functions as one network. He has allocated the Class B network **172.16.0.0** with mask **255.255.0.0** for the whole building.

Device	IP Address	Device	IP Address
Sales-PC1	172.16.1.10	Accts-PC1	172.16.2.10
Sales-PC2	172.16.1.20	Accts-PC2	172.16.2.20
Sales-PC3	172.16.1.30	Accts-PC3	172.16.2.30
Sales-PC4	172.16.1.40	Accts-PC4	172.16.2.40
Subnet Mask: `255.255.0.0` on all PCs			

Your Task

- Place two switches (one per floor), connect them together, and attach the four Sales PCs to the ground-floor switch and the four Accounts PCs to the first-floor switch.
- Assign IP addresses from the table above to all eight PCs.
- Verify that a Sales PC can successfully ping an Accounts PC across the inter-switch link.

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View showing both switches linked together with all eight PCs connected and IP labels visible.
2. A **ping result** in the Command Prompt of any Sales PC successfully reaching any Accounts PC, showing zero packet loss.

Note: Because all eight PCs share the same Class B subnet, the two switches can forward traffic between floors without any router involved.